DATA SCIENCE

Machine Learning Starter Pack

The algorithms, metrics and concepts that cover most entry-level data science interviews.

- **7 algorithms and when to use each**
- **The 99% accuracy trap**
- **60-day plan from zero**

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7 algorithms

You do not need 40. These cover most real tabular data problems.

- **Linear Regression**
  Predict a number. Next month sales.
- **Logistic Regression**
  Predict yes or no. Will this customer churn?
- **Decision Tree**
  Easy to explain to business. Clear if-then rules.
- **Random Forest**
  Many trees voting. Strong first model on most datasets.
- **XGBoost / LightGBM**
  Usually the best accuracy on tabular data.
- **K-Means**
  Group similar customers when there is no label.
- **KNN**
  Predict from the nearest similar records. Simple baseline.

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Pick the right metric

Accuracy alone can lie. Choose by what a mistake costs.

- **Precision**

  Of everything flagged, how much was right? Use when false alarms are costly, like spam.

- **Recall**

  Of all real cases, how many did we catch? Use when missing one is costly, like fraud or disease.

- **F1 score**

  Balance of precision and recall.

- **RMSE / MAE**

  Average error when predicting a number. MAE is easier to explain.

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"99% accurate. Why is that bad?"

A favourite interview question. Here is the full answer.

The trap

If 1% of transactions are fraud, a model that always says "not fraud" is 99% accurate and catches zero fraud.

What to check

Confusion matrix, recall and precision on the fraud class.

How to fix

Class weights, oversampling (SMOTE), or lowering the decision threshold.

```
RandomForestClassifier(class_weight="balanced")
```

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60 days from zero

Two projects at the end are worth more than five certificates.

- **Days 1-10: Python and Pandas**
  Load, clean, group, merge, plot.
- **Days 11-20: Statistics**
  Distributions, hypothesis testing, regression.
- **Days 21-45: Scikit-learn**
  The 7 algorithms, train-test split, cross-validation, metrics.
- **Days 46-60: Two projects**
  Churn prediction and customer segmentation, each with a business recommendation.

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